Exploratory Data Analysis (EDA)

SmartCharging Analytics – EV Charging Behavior Analysis

Student Information

Student Name: Dwij Vala
Student ID: 2505369
CRS: Artificial Intelligence
Course: Data Mining
School: Udgam School for Children

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1. Introduction

The rapid growth of electric vehicles has created increasing demand for efficient charging infrastructure. Understanding usage patterns of EV charging stations is important for improving station placement, pricing strategies, and operational efficiency.

This Exploratory Data Analysis (EDA) aims to understand the patterns present in the EV charging dataset before applying advanced data mining techniques such as clustering, anomaly detection, and association rule mining.

EDA helps identify trends, relationships between variables, and potential anomalies in the dataset.

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2. Dataset Description

The dataset contains information about EV charging stations and their usage characteristics.

Important features in the dataset include:

- Station ID – unique identifier for each charging station
- Latitude and Longitude – geographical location of the station
- Charger Type – type of charging connector available
- Charging Capacity (kW) – maximum charging power
- Cost (USD/kWh) – charging price per kilowatt hour
- Distance to City (km) – distance of the station from the nearest city
- Usage Stats (Average Users per Day) – average number of users charging daily
- Station Operator – company operating the station
- Renewable Energy Source – whether the station uses renewable energy

- Reviews / Ratings – customer feedback score

These attributes help analyze demand patterns and station efficiency.

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3. Data Preprocessing

Before conducting analysis, the dataset was cleaned and prepared.

The following preprocessing steps were applied:

- Handling missing values in the dataset
- Filtering invalid values where necessary
- Standardizing numerical features for machine learning algorithms
- Converting categorical variables into binary variables for association rule mining

These steps ensure that the dataset is suitable for analysis and machine learning models.

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4. Exploratory Data Analysis

4.1 Distribution of Charging Demand

A histogram was used to visualize the distribution of the average number of users per charging station.

Observation:

The distribution shows that most charging stations experience moderate usage levels, while a smaller number of stations have significantly higher demand.

This suggests that demand may be concentrated in specific locations.

4.2 Charging Cost vs Station Usage

A scatter plot was used to analyze the relationship between charging cost and station usage.

Observation:

Stations with lower charging costs generally show higher usage, indicating that pricing may influence user behavior.

However, some higher-cost stations also show strong demand, possibly due to their location or charging capacity.

4.3 Correlation Between Variables

A correlation heatmap was created to visualize relationships between numerical variables in the dataset.

Observation:

- Charging capacity and station usage show moderate correlation.
- Distance from city appears to influence station demand.
- Charging cost may have a negative correlation with usage.

These relationships help identify which factors impact station performance.

4.4 Geographic Distribution of Charging Stations

Using latitude and longitude data, stations were visualized on a map.

Observation:

Stations located closer to urban areas tend to show higher demand compared to stations located farther away from cities.

This indicates that location plays an important role in charging station utilization.

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5. Key Insights from EDA

The exploratory analysis revealed several important insights:

- EV charging demand varies significantly across stations.
- Charging cost may influence user demand.
- Geographic location is a major factor affecting station usage.
- Some stations display unusual patterns that may require further investigation.

These findings help guide the next stage of the project, which includes clustering analysis, anomaly detection, and association rule mining.

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6. Conclusion

The exploratory data analysis provided valuable insights into EV charging station behavior. The dataset shows clear patterns related to station usage, pricing, and geographic location.

These insights provide a strong foundation for applying advanced data mining techniques to further understand EV charging patterns and optimize infrastructure planning.